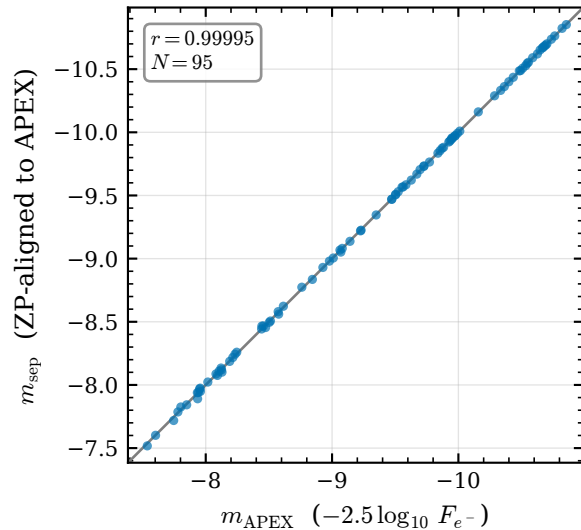
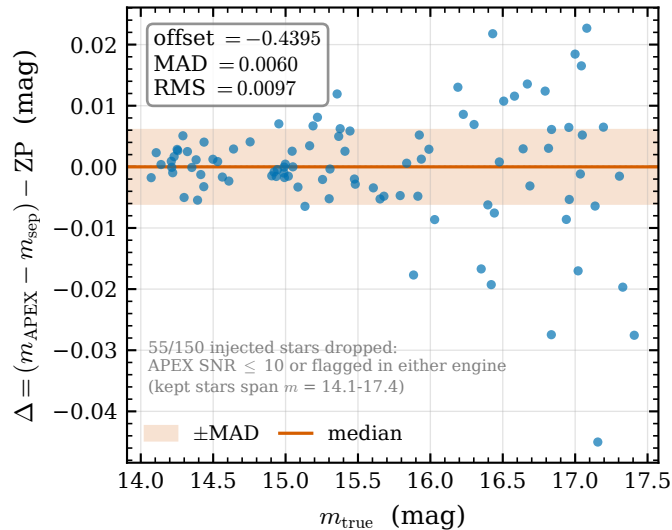


(a) two engines, same 95 stars: $r = 0.99995$ (b) residuals flat over $m = 14.1$ - 17.4 : MAD 6.0 mmag

Synthetic frame, no external data: noise model of apex.benchmark.synthetic_frame (seed 20260702 · 800² px · 150 isolated stars, $\text{sep} \geq 8 \times \text{FWHM}$ · Gaussian PSF FWHM 3.5 px · gain 1.5 e⁻/ADU · RN 5 e⁻ · sky 150 ADU · true $m = 14$ -20). Both engines integrate the same aperture ($r_{\text{ap}} = 3.5 \text{ px} = 1.0 \times \text{FWHM}$) at the same known centres: APEX phot_vectorized (local annulus sky) vs sep v1.4.1 Background + sum_circle (global sky). One median zeropoint alignment; no per-star fitting.